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B

The distance from the node to the end of the tube is $\frac{\lambda}{4}$. Since $\frac{\lambda}{4} = 0.17 \rightarrow \lambda = 0.68 \text{ m}$. The frequency to generate this stationary wave is $f = \frac{v}{\lambda} = \frac{340}{0.68} = 500 \text{ Hz}$

The wavelengths of $\frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \frac{7\lambda}{4}, \dots$ are equal to 0.17 m to obtain a next node at the same position.

The corresponding resonant frequencies would then be 500 Hz, 1500 Hz, 2500 Hz, 3500 Hz, 4500 Hz, 5500 Hz... etc.

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D

We cannot use Coulomb's Law because that is for point charges (or charge distributions with